

Whole-grain ready-to-eat oat cereal, as part of a dietary program for weight loss, reduces low-density lipoprotein cholesterol in adults with overw...

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OBJECTIVE: Weight loss and consumption of viscous fibers both lower low-density lipoprotein (LDL) cholesterol levels. We evaluated whether or not a whole-grain, ready-to-eat (RTE) oat cereal containing viscous fiber, as part of a dietary program for weight loss, lowers LDL cholesterol levels and improves other cardiovascular disease risk markers more than a dietary program alone. DESIGN: Randomized, parallel-arm, controlled trial. SUBJECTS/SETTING: Free-living, overweight and obese adults (N=204, body mass index 25 to 45) with baseline LDL cholesterol levels 130 to 200 mg/dL (3.4 to 5.2 mmol/L) were randomized; 144 were included in the main analysis of participants who completed the trial without significant protocol violations. INTERVENTION: Two portions per day of whole-grain RTE oat cereal (3 g/day oat β -glucan) or energy-matched low-fiber foods (control), as part of a reduced energy (approximately 500 kcal/day deficit) dietary program that encouraged limiting consumption of foods high in energy and fat, portion control, and regular physical activity. MAIN OUTCOME MEASURES: Fasting lipoprotein levels, waist circumference, triceps skinfold thickness, and body weight were measured at baseline and weeks 4, 8, 10, and 12. RESULTS: LDL cholesterol level was reduced significantly more with whole-grain RTE oat cereal vs control (-8.7 ± 1.0 vs $-4.3 \pm 1.1\%$, $P=0.005$). Total cholesterol (-5.4 ± 0.8 vs $-2.9 \pm 0.9\%$, $P=0.038$) and non-high-density lipoprotein-cholesterol (-6.3 ± 1.0 vs $-3.3 \pm 1.1\%$, $P=0.046$) were also lowered significantly more with whole-grain RTE oat cereal, whereas high-density lipoprotein and triglyceride responses did not differ between groups. Weight loss was not different between groups (-2.2 ± 0.3 vs -1.7 ± 0.3 kg, $P=0.325$), but waist circumference decreased more (-3.3 ± 0.4 vs -1.9 ± 0.4 cm, $P=0.012$) with whole-grain RTE oat cereal. Larger reductions in LDL, total, and non-high-density lipoprotein cholesterol levels and waist circumference were evident as early as week 4 in the whole-grain RTE oat cereal group. CONCLUSIONS: Consumption of a whole-grain RTE oat cereal as part of a dietary program for weight loss had favorable effects on fasting lipid levels and waist circumference. Copyright 2010 American Dietetic Association. Published by Elsevier Inc. All rights reserved.